

Dear Students:

In the MSK module there are Directed Learning Activity sessions which focus on genetics content. This material will be discussed during two clicker sessions in the MSK module which are called:

Inherited Disorders of the Musculoskeletal System and Triplet Repeat Disorders & Pharmacogenetics.

Required reading for ***Inherited Disorders of the Musculoskeletal System*** will entail:

1. Reading material from the Korf and Irons 4th edition textbook, and website as listed on this page.
2. Case studies which are part of this PDF which include 2 patients with Duchenne muscular dystrophy (DMD) and one patient with osteogenesis imperfecta (OI).
3. One (1) video that support OI and DMD - these are posted on Panopto.
4. A PDF of the PowerPoint slides for this video.

Directed readings from the Korf and Irons 4th edition textbook (note that these are brief descriptors set up as “break out boxes” and are very easy to digest.

Page 33	Charcot-Marie-Tooth disease (Clinical Snapshot 2.2)
Page 46-47	Marfan syndrome; also Figure 3.11
Page 46-47	Osteogenesis imperfect; also Figure 3.12, on page 48
Page 47	Achondroplasia; also Figure 3.13, on page 48
P 92-94 and 219 -	Alzheimer disease (on page 219, this is found on the top left hand column).
P 171-180	Neurofibromatosis - Note: focus on <i>NF1</i> .

On the website: <http://www.korfgenetics.com/factsheets.asp> (these are quick synopses of the disorders)

Go to [Neuromuscular disorders](#), and select:

- [41. Duchenne/Becker muscular dystrophy](#)

Go to [Cancer predisposition syndromes](#), and select:

- [18. Neurofibromatosis](#)

Go to [Skeletal/connective tissue](#), and select:

- [64. Osteogenesis imperfecta](#)
- [65. Marfan syndrome](#)
- [67. Achondroplasia](#)

Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

Muscular Dystrophy patient 1: A 5-year-old boy is brought to the physician because of weakness of upper and lower limb muscles. Due to the muscle weakness, he wants to be carried after walking a short distance. He also feels very sensitive to pain and is unable to climb up and downstairs on his own.

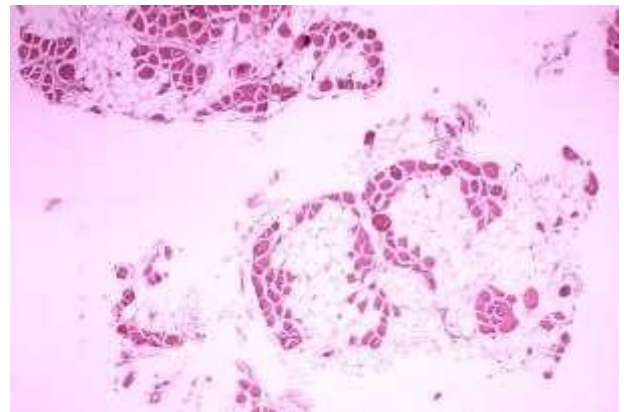
- **Past medical history:** significant for neonatal jaundice after birth. Muscle weakness is getting worse over time (progressive).
- **Family history:** Diabetes mellitus, hypertension, cardiomyopathy in the mother, and myocardial infarction in a maternal aunt.
- **Personal History:** He had a craving for indigestible things suggesting a neurological involvement
- **Physical Examination:** Affectionate, moody and biting his pencil. It was also observed that he was continuously moving from place to place in the consulting room. He had a positive Gower's sign, which is a description of the child helping himself to stand with upper extremities: first by rising to stand on his arms and knees and then "walking" his hands up his legs to stand upright. There was an apparent enlargement of his calf muscles – this is called pseudohypertrophy.

Investigations:

- Creatine kinase (CK-MM) level in the bloodstream was extremely high, 40,000 IU/L (normal is 25-200 IU/L)
- Electromyography (EMG) shows destruction of muscle tissues
- Genetic testing reveals a deletion in the X-linked *DMD* gene encoding the dystrophin protein.
- Muscle biopsy (followed by immunohistochemistry) confirms that dystrophin protein is absent (see fig. 2, lowest panel on page 2).

Fig. 1 Shows a Histopathology cross section of a biopsy from the gastrocnemius muscle of this patient: Note the extensive replacement of muscle fibers by adipose tissues.

The patient and the family were subsequently commenced on supportive and multidisciplinary treatments and the parents were referred for genetic counseling.



Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

Muscular Dystrophy patient 2: A 26-year-old man with slowly progressive weakness of the upper and lower limb muscles over the past 10 years comes to the physician. He was normal until the age of 16; after which he began to notice weakness of the lower limb muscles with difficulty climbing and rising from a sitting position. This slowly progressed to involve the upper limbs. He had, however, remained ambulatory until recently when he started using a wheelchair occasionally.

Family history: A 52-year-old maternal uncle has a similar condition.

Physical Examination:

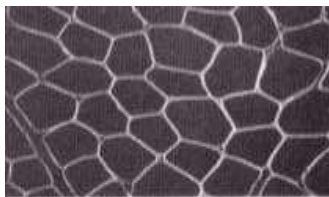
- Healthy looking, but anxious young man
- Mild enlargement of the calves and thinning of the thigh (pseudohypertrophy of the calf)

Investigations:

- Serum Creatine kinase level was 10,000 IU/L (normal 25-200 IU/L)
- Immunofluorescence shows reduced amount of Dystrophin protein (Fig. 2, middle panel)
- Targeted gene analysis shows a deletion mutation in the Dystrophin (*DMD*) gene.

Patient 2 was diagnosed with Becker muscular dystrophy. He was offered multidisciplinary supportive therapies and referred to a cardiologist.

Figure 2: Indirect immunofluorescence. An antibody to dystrophin protein is incubated with a histological slice from a muscle biopsy. A fluorescent conjugated secondary antibody is then applied. Exposure to the excitation wavelength specific for the conjugated fluorophore will reveal where this secondary antibody is, and an image may be obtained. Since the dystrophin protein is on the surface of the muscle fiber, each fiber is revealed.



The topmost panel shows a biopsy from a normal person with a strong signal of dystrophin protein outlining each sarcolemma of the muscle fibers.



The middle panel is that of patient 2 (Becker muscular dystrophy) which shows reduced abundance of dystrophin protein.



The bottom panel is that of patient 1 (Duchenne muscular dystrophy), with almost complete absence of dystrophin protein or severely reduced abundance.

Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

QUESTIONS:

1. Outline the clinical manifestations of Duchene and Becker muscular dystrophies.
2. What is the most common mutation that causes DMD? List the other (less common) types.
3. Both of these patients have a deletion in the same gene. Why is one more severe than the other?
4. What is the rate of new mutation for DMD? Compare the rate of new mutation in DMD to that for DMD.
5. Explain the reason for the high mutation rate for DMD.
6. What is Dystrophin protein? Explain its molecular functions.
7. Patient 2 was referred to the cardiologist. Give reasons for the referral.
8. Why do some female carriers do present with clinical manifestations of DMD? Explain the genetic basis for this.
9. What is Genetic Fitness?
10. Explain some differences (+/or similarities) between DMD and BMD with respect to the following headings:
 - (a) Disease severity
 - (b) Genetic fitness
 - (c) Type of mutation
11. Differentiate between new mutation and germ line mosaicism.
12. Is there a neurological function of the dystrophin protein?

Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

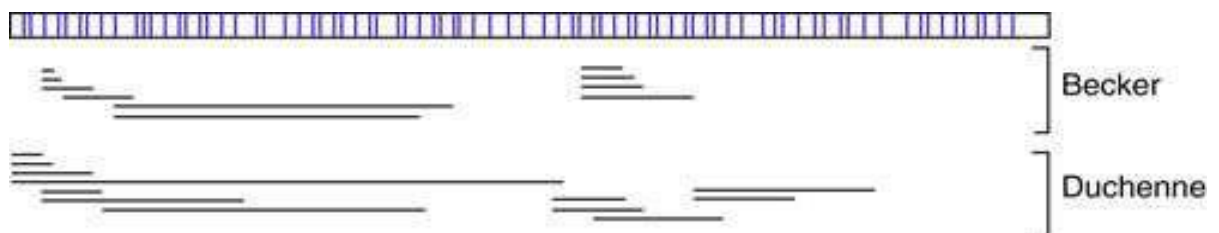
Discussion regarding the questions from the DMD cases

1. Patients with DMD usually present with:

- Marked and progressive weakness of upper and lower limb muscles starting at an early age; patient learns to walk but later develops clumsiness. The weakness starts with the large muscles of the pelvic girdle and is often obvious when he or she is asked to stand quickly from the lying position, where due to the weakness of the hip muscles, the patient uses the hands to help support the body as he or she rises (Gower sign)



- Calf muscle hypertrophy is caused by muscles growing to try to support weight, but most is fatty tissue.
 - Are wheelchair bound by the age of 12 years and typically develop cardiomyopathy by 14-18 years.
 - Males rarely live beyond their teens or early twenties hence genetic fitness is close to zero. Most common cause of death is often respiratory problems and cardiomyopathy.
 - Female carriers may present with symptoms, and this depends mainly on the degree of X-inactivation.
2. About 2/3 of all DMD causing mutations are caused by a deletion mutation. Patients with most often have a deletion that leads to a change in the reading frame of the coding sequence. Very large deletions may also lead to DMD. Frame shifts often result in premature termination codon formation and a truncated protein which is non-functional.



Less common mutations will include:

- Duplication events
- Single base pair mutations (missense, nonsense)
- Mutation of a small number of base pairs
- Small insertions or deletions (frameshifts)

These other DMD causing mutations in dystrophin are more difficult to detect. They may require direct sequencing and analysis, hence are more expensive.

Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

3. DMD is mostly caused by deletions that lead to loss of proper reading frame – causing very severe disease. In contrast, in BMD, the correct reading frame is likely to be maintained after the deletion or duplication (reading frame not affected). In BMD, there is some deleterious effect due to the mutation; it is just not as drastic as in DMD.
4. The rate of new mutation in DMD is high. About 1/3 of cases are predicted to result from new mutation whereas in BMD, only about 10% of cases represent new mutation.

Most of the De-novo large deletion mutations occur during oogenesis. If an ovum with this mutation is fertilized by a normal 23Y sperm, this male child will have DMD; If the ovum is fertilized by a normal 23X sperm, the female child will be a carrier, unless skewed X-inactivation occurs, and she becomes a manifesting heterozygote.

Most of the De-novo point mutations are formed during spermatogenesis. If this sperm fertilizes a normal ovum, it will result in a carrier female.

5. The high mutation rate for DMD can be partly explained by the fact that the locus is a large target for mutation. The most remarkable feature of the DMD gene is its large size. It is massive, and by far the largest gene in the human genome.
6. Dystrophin is a structural protein that anchors a large protein complex at the cell membrane;
 - Dystrophin links the actin cytoskeleton with a protein complex that is associated with the sarcolemma membrane.
 - Dystrophin is also required to position an assortment of other proteins in the complex so that they all function correctly, as seen in the neuromuscular junction for proper acetylcholine clustering during development.
 - Without this structural support, the cell membrane becomes permeable, leads to damage to the muscle fibers (sarcolemma breaks down, calcium influx occurs, there is phospholipase activation, oxidative cell injury, serum creatine kinase leaks out of the muscle). It can also lead to myonecrosis.
7. Patient #2 is male, and he has Becker muscular dystrophy. The dystrophin protein is also important cardiac muscle which is likely to be deficient for this protein. This means that he is at increased risk of having a cardiac problem in his lifetime, hence the referral.
8. About half of all women who are carriers of a DMD mutation will present with clinical manifestations of DMD as they age. Consider a woman who is a carrier of DMD, and she has normal X-inactivation; if this is exactly half, then half of the heart tissue will have a deficiency of the dystrophin protein. Skewed X-inactivation might increase a woman's risk for DMD if the X-chromosome that contains the normal DMD gene is inactivated in a high percentage of cells, and the X-chromosome that harbors a mutation in DMD remains active. Higher degrees of unequal X-inactivation, will lead

Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

to more severe DMD. In exceedingly rare instances, females have been described with DMD due to events such as X:autosome translocations or Turner syndrome (45X).

9. Genetic fitness refers to the ability to pass on one's genes to offspring, the ability to reproduce:
- **Zero genetic fitness (genetic lethal):** The affected individual is unable to reproduce. Examples include DMD and Tay Sachs disease.
 - **Reduced genetic fitness:** Some individuals who have the disorder are still able to reproduce. An example may be any disorder which has a negative impact on a person's ability to have a child.
 - ***Normal genetic fitness:** Examples include Huntington disease (because onset typically occurs after childbearing years), and adult-onset polycystic kidney disease.

Note that genetic fitness can change depending on the treatment available. For instance, a person who has appropriate dietary treatment for PKU will lead a normal life and have normal genetic fitness. Whereas an untreated PKU patient will have severe intellectual impairment and is unlikely to reproduce – very low genetic fitness.

10. Explain some differences (+/or similarities) between DMD and BMD with respect to the following headings: 1] Disease severity; 2] Genetic fitness; 3] Type of mutation.

	Duchene Muscular Dystrophy	Becker Muscular Dystrophy
Disease Severity	Very severe Onset at early age Wheelchair bound by late teens	May have same features as DMD but less severe Age of onset usually after 7 years old
Genetic Fitness	Near zero reproductive fitness. Often lethal before the age of 30 yrs. Due to the pathology, very few are able to reproduce.	Relatively high Up to 70% of normal
Type of Mutation	Large deletions Duplication Single base pair mutation, small insertion or deletion (which lead to frame shifts) Mutation of small number of base pairs	Deletions or duplication which often result in in-frame Changes to exons, That there are no changes in the reading frame of the codons

11. New mutation and germ line mosaicism are sometimes difficult to understand.
- If the disorder occurred due to a new mutation during oogenesis, there will be an affected child with no family history of the disorder. Recurrence risk would be

Muscular Dystrophy Case: Genetics Cases for Clicker Questions and Discussion in Class

very low, making it highly unlikely to see more than 1 affected child or individual in the family.

- It is well known that increased age is a risk factor for new mutation in fathers. Consider a man who passes on a new DMD mutation that occurred during spermatogenesis. The man is XY, so he can ONLY pass this mutation on to a daughter, so he can only create a carrier daughter – he may not pass this mutation on to a son. Therefore, a new mutation that occurs during spermatogenesis will typically only become clinically evident in his grandchildren. (draw the pedigree to show this)
- Germ line mosaicism is the presence of more than one cell line in the gametes in an otherwise normal parent and is the result of a mutation during embryonic development of that parent. In germ line mosaicism, the mutation occurs during germ cell development and may involve multiple sperms and egg cells. This is important because an individual with germ line mosaicism is at increased risk of having more than one affected child in spite of not being affected with the trait.

12. There is evidence that the dystrophin protein has a neurological function.

One of the functions of the dystrophin complex is its involvement in the positioning of proteins in higher order complexes so that they can function correctly. For example, the dystrophin complex is required at the neuromuscular junction for proper acetylcholine clustering during development.

There are suggestions that dystrophin also plays a role in brain development.

OI Case: MSK Module for Genetics Cases Discussion in Class

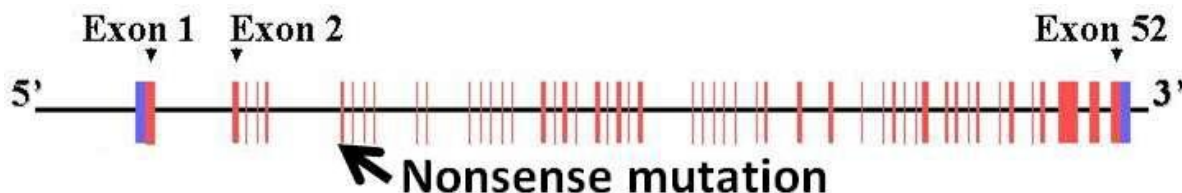
OI Patient 1 is a 3-year-old girl in the pediatric ward who is seen as a follow-up to an ER visit for a humerus fracture. The parents had no idea when or how it happened; they reported that when they held her by the upper arm, she indicated pain. When she was 2-years-old she had a tibial fracture. She had a limp, and parents brought her to the ER where X-ray showed the bone break. At that time, it was diagnosed as a Toddler's fracture. Now, the family is being investigated for child abuse. A careful history reveals the following. The father is very active and is involved in many sports clubs. The mother is the exact opposite, she likes to read, and does not engage much in physical activity, she never has. The child has blue sclera (see below). The mother notes that her father also had this blue coloration in his eyes, but he died relatively young in a car crash. The mother remembers being told that her father had many broken bones when he was young; this was attributed to him being a “wild child”. Patient #1 doesn't seem to have any bruises, scars or any other marks on her body that might be consistent with child abuse. A skeletal survey is requested to look for occult fractures in different stages of healing. The child has near-normal stature based on pediatric growth chart but seems to be smaller than expected when compared to her siblings. A DNA analysis is suggested with the hope of identifying a genetic reason for the fragile bones of this patient.



Genetic analysis shows the following information:

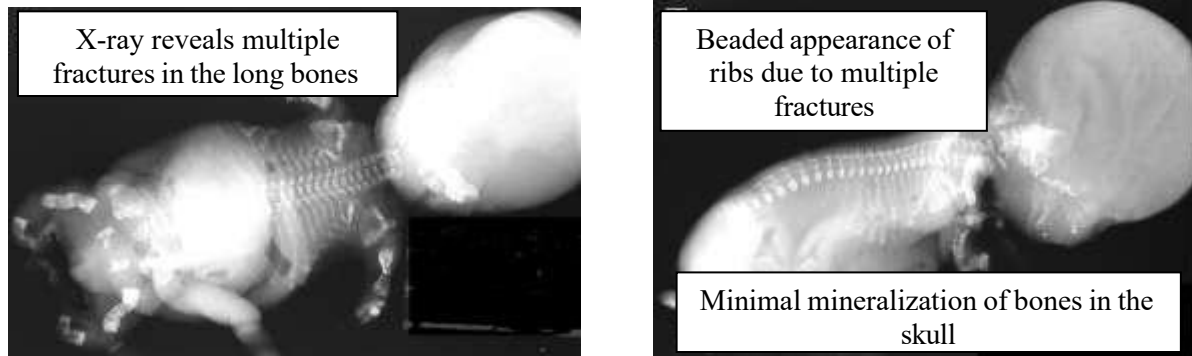
- There are no paternity issues in the family
- Patient #1 has a “nonsense” mutation near the 5' end of one copy of the *COL1A1* gene. Note from the cartoon that this mutation occurs in of the early exons of the gene.
- This same alteration (mutation) of the genetic code is also present in the mother
 - Recall that a nonsense mutation creates a stop codon, so in this case, the protein encoded by the allele inherited from the mother is likely to never be produced
- The father's *COL1A1* gene sequence is normal, on both copies
- The diagnosis of classic non-deforming OI with blue sclera is made. Note that the old term for this disorder was OI Type 1.

Cartoon of *COL1A1* gene showing position of nonsense mutation in patient #1 is shown. The *COL1A1* gene is 18 kilobases in size and is composed of 52 exons. This gene encodes a 1464-amino acid pro α 1 chain that will be used to construct type I collagen



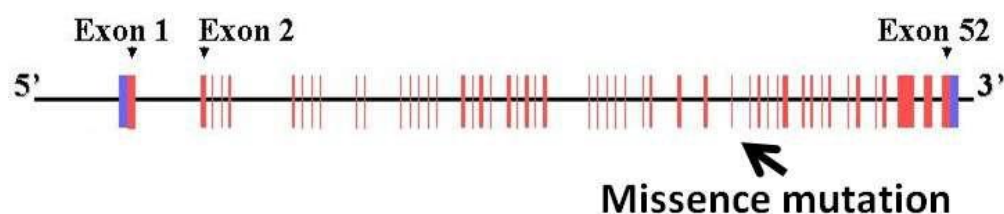
OI Case: MSK Module for Genetics Cases Discussion in Class

OI Patient 2: A 49-year-old man visits a genetic counselor with his 34-year-old wife because she had a spontaneous abortion late in the third trimester of a recent pregnancy. This would have been their second child. The first child of this couple is now eight years old and is in good health. Post-mortem X-ray of the abortus revealed multiple fractures in the long bones and ribs associated with generalized demineralization of all bones in the fetus. Based upon the perinatal lethality, and the post-mortem investigation, a diagnosis of Perinatally lethal osteogenesis imperfecta type II was made (note, this used to be called OI type II).



After a counseling session and obtaining informed consent from both parents, tissue from the abortus was obtained, DNA was isolated, and genetic analysis was conducted. Whole exome sequencing was conducted using next generation sequencing systems, and the following information was revealed:

- There are no paternity issues in the family
- The fetus had a mutation in the *COL1A1* gene
 - This is a missense mutation, as it is a single base pair change which results in a Gly to Val substitution in the *COL1A1* protein
 - The position of this mutation is shown below in the cartoon structure of the *COL1A1* gene



- This mutation was not present in either parent
- This mutation had never been described before in the medical literature, so at this point it wasn't clear how to "call" the mutation. The first geneticist wanted to call it a "**Likely Pathogenic Mutation**", and a second geneticist wanted to call it a "**Mutation of Unknown Clinical Significance**"
- Based on the nature of the mutation, its position in the gene, and the features observed in the aborted fetus, the parents were informed that this mutation **might** explain the disorder.
 - To confirm this diagnosis the synthesis of mature collagen would have to be studied *in vitro*
 - About a year later it was found that when this gene was expressed *in-vitro*, the mutation indeed interferes with normal formation of collagen, and in a sense, it

OI Case: MSK Module for Genetics Cases Discussion in Class

dominantly disrupts the formation of collagen (hence the mutation is defined as “dominant negative”).

- At this point the first geneticist called the second and said, “I told you so”.

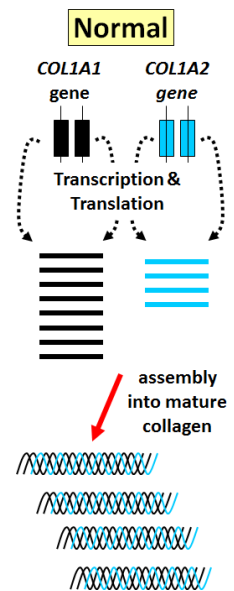
Questions

1. What is the normal structure of collagen and how is it normally formed?
2. What are the ranges of OI, and how do the clinical findings in these cases match to types of OI classifications?
3. Regarding the patient in case 1:
 - a. What is the effect of the mutation in this patient?
 - b. What is a nonsense mutation?
 - c. How can a splice site mutation also lead to the formation of an early stop codon?
 - d. Why does a frameshift mutation often lead to the formation of a stop codon soon after the mutation?
4. Regarding the patient in case 2:
 - a. What is the effect of the mutation in this patient?
 - b. Consider the effect of the amino acid change because of the mutation. How would this alteration change the structure of collagen? Is this considered a dramatic change?
 - c. Why is the disorder more severe in case 2 than in case 1?
5. What important genetic principles are illustrated by *COL1A1/2* related OI
 - How can OI cases exemplify the genetic principle of locus heterogeneity?
 - How do OI cases exemplify the genetic principle of allelic heterogeneity?
 - How does this allelic heterogeneity affect the severity of disease in affected individuals (genotype / phenotype correlation)?
 - Why is the term “haploinsufficiency” used to describe the OI in patient #1?
 - Why is the term “dominant negative” used to describe the OI seen in patient #2 (the perinatal lethal case of OI)?
 - What is the relationship between genetic fitness and *de novo* mutations?
6. What are some other human disorders which are classified as dominant negative?

Commentary:

1. Normal collagen is a tightly wound triple helix comprised of three procollagen peptide chains. The ratio of these procollagen peptides is 2:1. Two of the procollagen peptides are derived from the *COL1A1* gene, and the other procollagen peptide is synthesized from the *COL1A2* gene.

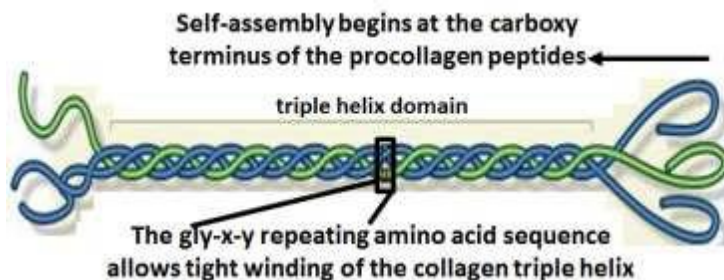
Since humans only have two copies (alleles) of these genes, it follows that twice as much synthesis occurs from the *COL1A1* gene than from the *COL1A2* gene. This may be explained by increased transcription and translation of the *COL1A1* gene, leading to twice as much protein product produced when compared to *COL1A2*.



The amino acid sequence of the procollagen peptides is remarkable in that there is a long repeat of composed of the three amino acids [glycine-x-y]. The presence of glycine, which holds the smallest R-group of all of the amino acids is significant as it allows tight packing of the triple helix. The -x and -y amino acids have a high proportion of proline and lysine, many of which are post-

translationally hydroxylated. This hydroxylation modification facilitates covalent crosslinking of the collagen which increases the stability of the mature protein. These hydroxylation reactions require ascorbic acid (vitamin C) and a deficiency of this vitamin leads to scurvy due to decreased stability of collagen fibers (recall from biochemistry).

After synthesis in the cytoplasm of the fibroblast and post-translational hydroxylation of the proline and lysine residues (vitamin C dependent), the procollagen chains are exported into the extracellular matrix where they assemble into collagen. Assembly begins from the C-terminus of the protein and then proceeds to the N-terminus of the procollagen peptides.



The final output is the production of four collagen triple helix proteins - each composed of three procollagen chains.

COL1A1/2-related OI is characterized by fractures with minimal or absent trauma due to a defect in collagen formation. Clinical presentations are spread from nearly asymptomatic individuals (consider the mother of case #1 Samantha L. Jackson) to perinatal lethality (Bruce W., in case #2). Between these two extremes a continuum exists: from severe skeletal deformities, mobility impairments, and very short stature to individuals with a mild predisposition to fractures, normal stature, and normal life span. Problems with tooth formation (dentinogenesis imperfecta), and progressive hearing loss also may accompany the disorder in addition to other symptoms that rely on proper collagen formation.

2. Current classification of *COL1A1/2* related OI types are now referred to as follows:

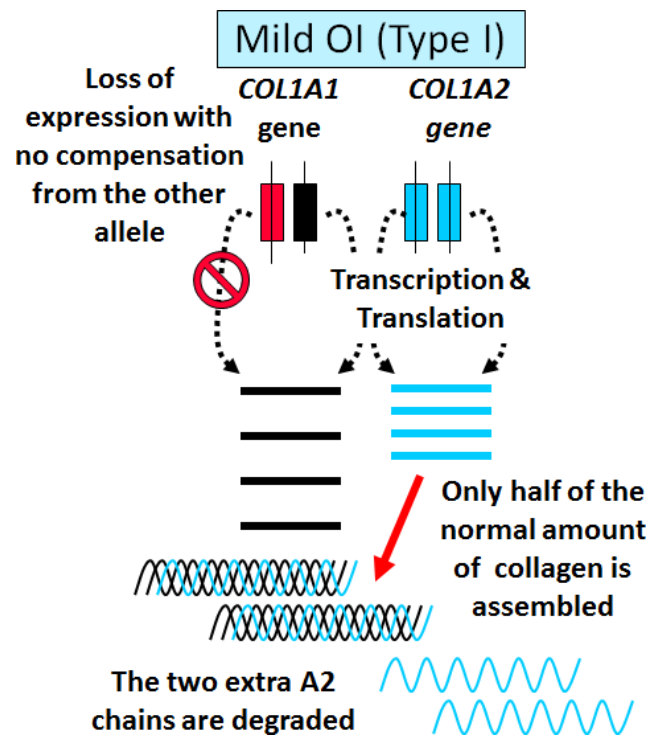
- Classic non-deforming OI with blue sclera (OI type I): - this is predisposition to fractures, non-deforming, & with blue sclera (patient in case #1). Typically, these are a result of loss of expression (haploinsufficiency) from one allele of *COL1A1*.
- Perinatal lethal OI (OI type II): due to severe defects in collagen formation (patient in case #2). Typically described as an example of severe dominant negative.
- Progressively deforming (OI type III). Not discussed in this case
- Common variable OI with normal sclera (OI type IV) not discussed in this case

Type II, Type III and Type IV OI are typically described as dominant negative. Type II is the most severe (perinatal lethality) – other types of OI are less severe than Type II.

It should be noted that these classifications represent a continuum and the features of the four *COL1A1/2*-related OI types often overlap. Also, the clinical features may change as the patient ages, and it is not always easy to categorize the extent of the clinical disorder. **For this case, and this course,**

We limit our discussions to Type I and Type II OI.

All of the OI disorders described in these cases may also result from mutations in the *COL1A2* gene. In this series of cases, we focus on the *COL1A1* gene to be consistent in our examples and comparisons. This is a good example of locus heterogeneity.



3. Classic non-deforming OI with blue sclera (OI type I): (case 1) results from the loss of expression from the *COL1A1* gene causing an imbalance of procollagen fibers resulting in decreased production of normal collagen.

The patient with Classic non-deforming OI with blue sclera (OI type I) produces only about ½ the normal amount of collagen – this leads to the clinical manifestations in the patient. However, the collagen that actually gets produced is indeed perfectly **normal**.

A nonsense mutation is a point mutation that results in the formation of a stop codon in the mRNA. In the case of a nonsense mutation, the mRNA is usually processed normally (there are some exceptions) and the mRNA may be of normal length. However, since a stop codon occurs before it is normally supposed to, a truncated protein is produced during translation on the ribosome. This abnormal peptide is usually degraded and has no cellular function.

There are other genetic events that can lead to the formation of a stop codon in the incorrect place. For instance, a frame shift mutation results from the insertion or deletion of nucleotides. If the insertion/deletion is of a number other than a multiple of three, all codons after the mutation will specify a different amino acid, and by pure chance, an inappropriate stop codon will occur to halt translation.

Another way a premature stop codon may occur is by splice site mutation. Splice donor / splice acceptor junctions that join exons and allow removal of introns are not necessarily in frame. That is, the end of one exon does not always end with the triplet of a codon. Sometimes, the codon triplet begun at one exon must be finished by appropriate splicing to the next exon. If an exon is deleted, and the pre-mRNA is spliced to the next available exon, loss of reading frame may (or may not) occur – depending on the codon arrangement.

4. Depending on the location of the mutation, a point mutation in *COL1A1* may be more severe than a mutation that results in loss of expression. Perinatal lethal OI (OI type II) results from a mutation in the *COL1A1* gene which destabilizes the collagen triple helix.

The specific mutation described in case 2 is a mis-sense mutation which results in the substitution of a valine residue in place of a glycine. Glycine is found at approximately every third position in collagen as it has a very small R- group (only a hydrogen atom), whereas valine has a bulky branched aliphatic chain. The presence of this bulky side chain destabilizes the collagen triple helix, causing “brittle bones”. This glycine to valine change is considered dramatic, since the structural alteration is quite severe.

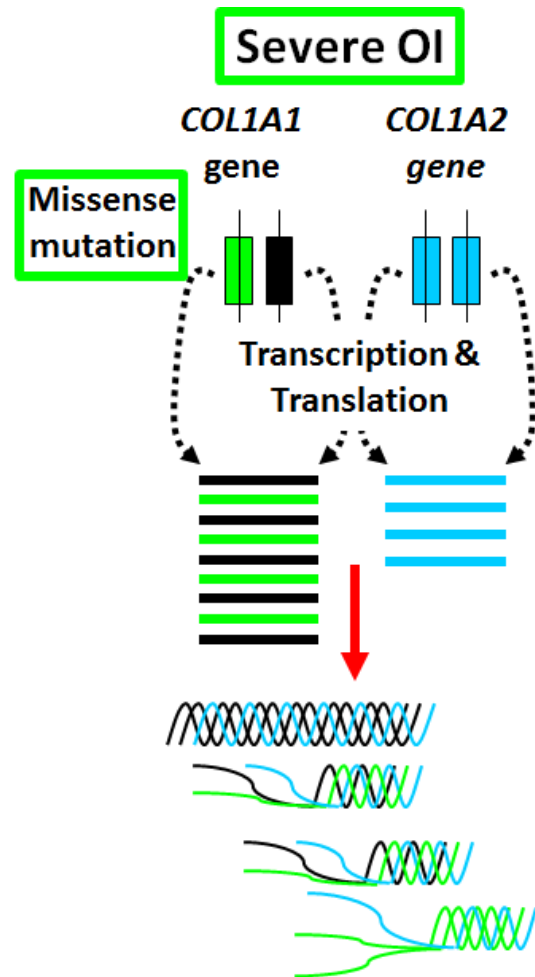
Compared to case 2, the patient in case 1 has a less severe phenotype because the collagen that is actually produced in her cells is normal – the problem is that only half the normal amount is made.

To summarize: in classic non-deforming OI with blue sclera (OI type I), there is a 50% reduction of normal collagen (this is haploinsufficiency). On the other hand, in Perinatal lethal OI (type II), only 25% of the collagen that is made is normal, and 75% is created with an abnormal protein which destabilizes the final collagen structure (this is dominant negative). Depending on the mutation, there is a range of severity from lethal (Type II, to non-lethal – Type III, Type IV, etc). Perinatal lethal OI (type II), being the most severe.

6. OI is a very rare condition. The most common genes associated with OI are: the *COL1A1*; and *COL1A2*. Mutations in these genes are typically inherited in an autosomal dominant fashion. There are many other genes associated with OI, and some are inherited in autosomal recessive pattern. These other types of OI are exceedingly rare and are not discussed in this course.

Autosomal dominant *COL1A1/2* related Osteogenesis imperfecta (OI) illustrates many important principles in medical genetics. These principles may be reviewed from your notes and from the textbook.

- Since mutations in two different genes, at two distinct chromosomal locations contribute to the same phenotype *COL1A1/2* related OI exhibits **locus heterogeneity**.
- There are many different mutations which may affect the structure or ability to form collagen, therefore *COL1A1/2* related OI also exhibits **allelic heterogeneity**.
- Although generalizations are difficult in that there are always exceptions, a few **genotype /**



phenotype examples can be made with *COL1A1/2* related OI. For instance, mutations which result in loss of expression typically result in classic non-deforming OI with blue sclera (OI type I), since some normal collagen is still produced. Mutations which destabilize the collagen triple helix are typically more severe ranging from perinatal lethal type II OI to type III and type IV.

- Type II, (also Type III, and Type IV) OI nicely illustrate the genetic principle of **dominant negativity**. In the case of dominant negative OI the mutation causes one subunit of the mature protein to destabilize the final collagen structure. In turn, this leads to weaker bone matrix
- Type I OI is best explained by **haploinsufficiency** as it is simply the result of loss of expression, and a decreased abundance of mature collagen
- As an autosomal dominant disorder exhibits lower genetic fitness, the probability that the disorder is caused by a **de novo** mutation increases. [NOTE: This is not the case for autosomal recessive disorders since most disease alleles are masked by the functional allele. Another way to put this is that for AR disorders, most disease alleles are already in the population but are hidden in heterozygotes.]

7. Another classic example of a dominant negative disorder is Marfan syndrome.

Final notes: There are always exceptions: there are examples of OI being inherited in an autosomal recessive pattern, but this is esoteric, complex, and exceedingly rare. It is not discussed in our course.

Have you ever seen the movie called *Unbreakable*? It is a very slow-moving film starring Bruce Willis and Samuel L. Jackson. If you can stay awake through it, you may notice that it is about a comic book hero (Bruce Willis) and a patient with OI (Samuel L. Jackson).

References: Gene Reviews, available at www.genereviews.org, material from previous modules, Lippincott's Illustrated Review's – Medical Biochemistry, and Korf and Irons, 4th edition.